Online Exam Portal – Test Cases

Functional Requirement Test Cases

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
TC_01	Project Setup	Verify the application starts and required pages load.	Project is installed/configured as described by the project setup.	1. Start the application. 2. Open the main page. 3. Navigate to login.	Configured application	Application starts and the expected entry pages load without an error.		
TC_02	User Login Authentication	Verify login with valid credentials.	A registered user account exists.	1. Open login page. 2. Enter valid credentials. 3. Submit login form.	Valid registered credentials	User is authenticated and redirected to the appropriate page for their role.		
TC_03	User Login Authentication	Verify invalid credentials are rejected.	Login page is available.	1. Enter an invalid username/email or password. 2. Submit the form.	Invalid credentials	Access is denied and an appropriate error message is shown.		
TC_04	Role-Based Access Control	Verify users can access only functions allowed for their role.	Test accounts exist for the roles implemented in the project.	1. Log in with each test role. 2. Open permitted pages. 3. Attempt to open a restricted page/action.	Student and authorized staff/admin accounts, where applicable	Permitted actions are available; restricted actions are denied.		
TC_05	View Available Exams	Verify a student can view available examinations.	Student is logged in; an examination is available to that student.	1. Open the student exam area. 2. Review the available examination list.	Published/available examination	Available examinations are displayed with the information provided by the portal.		
TC_06	Create Examination	Verify an authorized user can create an examination.	An authorized account is logged in.	1. Open examination management. 2. Enter the required exam details. 3. Save the examination.	Valid examination details	Examination is saved and appears in the relevant management/list view.		
TC_07	Question Bank	Verify questions can be added to the question bank.	An authorized account is logged in and question-bank functionality is available.	1. Open question bank. 2. Enter question and required answer/options. 3. Save.	Valid question and answer data	Question is saved and appears in the question bank.		
TC_08	Question Bank	Verify an existing question can be edited.	At least one question exists and the user has edit permission.	1. Select a question. 2. Edit its content or options. 3. Save and reopen it.	Updated question text/options	Saved changes are displayed and the question remains available.		
TC_09	Question Bank	Verify a question can be deleted by an authorized user.	At least one question exists and the user has delete permission.	1. Select a question. 2. Choose Delete. 3. Confirm if prompted.	Existing test question	Question is removed from the question bank, subject to the portal's implemented rules.		
TC_10	Attend Examination Online	Verify a student can start and answer an available examination.	Student is logged in and has access to an available examination.	1. Open the exam. 2. Start the attempt. 3. Select answers for questions.	Available exam with multiple questions	Exam questions are displayed and answers can be entered.		
TC_11	Examination Countdown Timer	Verify the countdown timer reflects the configured exam duration.	An examination attempt with a configured duration is active.	1. Start the exam. 2. Observe the timer at intervals.	Configured exam duration	Remaining time decreases during the attempt and is displayed consistently.		
TC_12	Automatically Save Exam Answers	Verify answers are saved while the student is taking an exam.	An exam attempt is active and answer-saving behavior is implemented.	1. Answer a question. 2. Navigate to another question. 3. Return and verify the first answer.	Selected answer	The previously selected answer is retained according to the implemented save behavior.		
TC_13	Navigate Between Examination Questions	Verify students can move between questions.	An exam attempt with at least two questions is active.	1. Move to the next question. 2. Move to the previous question. 3. Check displayed question and selected answers.	Exam with multiple questions	Navigation displays the correct questions and does not mix up answers.		
TC_14	Restrict Exam Access Before Scheduled Time	Verify access is restricted before an exam's scheduled start.	The exam has a configured schedule and access restrictions are implemented.	1. Log in before the scheduled start. 2. Attempt to open/start the exam.	Exam with a future start time	The student cannot start the exam before the allowed time and receives suitable feedback.		
TC_15	Automatically Submit Exam When Time Expires	Verify the configured timeout behavior at exam expiry.	An exam attempt with a short duration is active.	1. Start the exam. 2. Allow the timer to expire. 3. Observe the attempt status.	Short-duration exam	The attempt follows the implemented expiry behavior; if auto-submit is supported, the attempt is submitted automatically.		
TC_16	Automatically Evaluate Objective	Verify objective answers are scored	An objective-question exam with known	1. Submit an attempt with known correct and	Known answer key and scoring	The score matches the configured answer key		

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	Questions	against the configured correct answers.	answers and marking rules is available.	incorrect answers. 2. View the score if available.	rules	and marking rules.		
TC_17	View Examination Results	Verify an authorized student can view their examination result when available.	A submitted attempt exists and result visibility is enabled.	1. Open the results area. 2. Select the submitted examination.	Submitted exam attempt	The correct result for that student and examination is displayed when permitted.		
TC_18	Student Performance Reports	Verify a performance report can be generated or viewed.	Relevant student examination data exists and the account has permission.	1. Open student performance reports. 2. Select the relevant student/exam if required. 3. View the report.	Existing exam results	The report displays the performance information supported by the portal.		
TC_19	Examination Statistics	Verify examination statistics are displayed from available attempt data.	Examination attempts or result data exist and statistics functionality is available.	1. Open examination statistics. 2. Review the displayed values.	Existing attempt/result data	Displayed statistics correspond to the available examination data.		
TC_20	Secure User Credentials and Examination Data	Verify protected credentials and exam data are not exposed to unauthorized users.	Application is running and test accounts are available.	1. Log in as a user without the required permission. 2. Attempt to access protected user/exam data.	Unauthorized account	Protected data is not exposed and unauthorized access is denied.		
TC_21	Data Consistency During Network Loss	Verify exam answers/data remain consistent after a network interruption, if supported.	An exam attempt is active and the test environment can simulate network loss.	1. Enter an answer. 2. Simulate temporary network loss. 3. Restore connection and inspect the attempt.	Selected answer and temporary connection loss	The portal follows its implemented recovery behavior without silently corrupting or duplicating saved answers.		
TC_22	Backup and Recovery of Exam Data	Verify the configured backup/recovery procedure preserves examination data.	A backup/recovery mechanism and test environment are available.	1. Create or identify test exam data. 2. Run the documented backup/recovery procedure. 3. Verify recovered data.	Test exam, questions and attempt data	Recovered data matches the data covered by the configured backup/recovery process.		
TC_23	Maintain Activity Logs	Verify supported user/exam actions are recorded in activity logs.	Logging is enabled and the tester can access logs.	1. Perform a supported action such as login or exam creation. 2. Inspect the activity log.	Known user action	The expected action is logged with the details supported by the implementation.		
TC_24	Responsive and User-Friendly Interface	Verify key portal pages remain usable at common screen sizes.	Portal is available in a supported browser.	1. Open login, exam list and exam attempt pages. 2. Check at desktop and narrow/mobile viewport sizes.	Desktop and narrow viewport	Key controls and content remain readable and usable without major layout breakage.		

Execution note: Fill Actual Result and Test Result after running each case. Items involving scheduling, auto-save, auto-submit, network recovery, backup/recovery, logs, reports or statistics must be validated against the implemented project workflow; do not mark Pass without execution.

Reference note: Modules and feature names follow the provided backlog screenshots. The duplicate backlog entry for “US05 - Add Questions to Question Bank” is represented once as a single feature.